

LuxStudio

Simple Project Requirements Document

Project purpose

LuxStudio is a video editing and AI content-generation application designed to help users turn long videos into short, polished, social-media-ready clips. The system focuses on automating repetitive editing tasks such as removing silence, generating captions, finding useful clips, applying branding and creating social-media text.

1. The Problem

Editing long videos into multiple short clips can take a lot of time. Users often have to manually remove silent sections, create captions, search for interesting moments, add branding and write descriptions.

LuxStudio will reduce this manual work by combining these tasks into one focused application.

2. Project Goal

- Reduce the time needed to edit long videos.
- Turn one long video into several short clips.
- Automatically remove unnecessary silence.
- Generate editable captions.
- Use AI to suggest the best short clips.
- Create titles, summaries, descriptions and hashtags.
- Keep branding consistent.
- Export videos ready for social media.

3. Target Users

User	What they need
Video Editor	Edit videos, remove silence, create captions and export clips.
Content Creator / Speaker	Quickly turn long recordings into short, shareable content.
Administrator / Brand Manager	Manage branding, templates and reusable settings.

4. Main User Workflow

Step 1: Import a long video or audio recording.

Step 2: Edit using a timeline: cut, trim, split, delete, reorder or merge clips.

Step 3: Remove Silence automatically and close the gaps.

Step 4: Generate Captions from the spoken audio and correct any mistakes.

Step 5: Generate Clips using AI to suggest complete and interesting short moments.

Step 6: Apply Branding and choose a caption style.

Step 7: Generate Social Content such as a title, summary, description and hashtags.

Step 8: Review and Export the final clips.

5. Main Features

Project and Media Management

- Create, open, save and close projects.
- Import common video and audio formats.
- Show basic information such as video length and resolution.
- Automatically save work and recover projects after an unexpected close.
- Never overwrite the original source file.

Timeline Editing

- Display video, audio and captions on a timeline.
- Split, trim and delete clips.
- Reorder and merge clips.
- Zoom and navigate long recordings.
- Play, pause and seek through the video.
- Undo and redo editing actions.

Automatic Silence Removal

- Detect silent or nearly silent sections.
- Allow the user to adjust silence settings.
- Let the user review detected sections.
- Remove selected silent sections.
- Automatically close the gaps after removal.

Automatic Captions

- Convert spoken audio into text.
- Synchronize captions with the video.
- Allow users to correct transcription mistakes.
- Allow caption text and position to be edited.

Caption Templates

- Provide three starting caption styles.
- Allow changes to font, size, text colour and highlight colour.
- Support simple caption animation.

- Keep the selected style consistent across generated clips.

AI Clip Generator

- Analyse long recordings and identify useful short clips.
- Look for strong openings, complete ideas, clear teaching and memorable moments.
- Generate several clip suggestions for the user to review.
- Show the clip title/topic, start time, end time and duration.
- Allow the user to open and edit any suggested clip.
- Allow clip suggestions to be generated again.

AI Summary and Social Copy

- Generate a title.
- Generate a short summary and key points.
- Generate a social-media description.
- Generate relevant hashtags.
- Allow the generated text to be edited and copied.

Branding and Settings

- Store a logo, organisation name and other reusable details.
- Allow settings to be changed without changing the application's source code.
- Apply branding when enabled.

Export

- Export vertical 9:16 videos.
- Primary target: 1080 × 1920 MP4.
- Export individual clips.
- Support batch export as a future improvement.
- Show export progress and clear error messages.

6. What Is Included in the First Version (MVP)

Priority	Feature	Version
High	Import, preview and basic timeline editing	MVP 1
High	Automatic silence detection and removal	MVP 1
High	Automatic transcription and captions	MVP 1
High	Three editable caption templates	MVP 1
High	9:16 MP4 export	MVP 1
Medium	AI clip suggestions	MVP 2
Medium	AI summary, descriptions and hashtags	MVP 2
Medium	Branding and organisation settings	MVP 2
Later	Batch export and advanced clip scoring	Future
Later	More social-media export formats	Future
Later	Cloud projects and team collaboration	Future

7. User Experience Requirements

- The application should be easy to understand without advanced technical training.
- The main actions should be easy to find: Import, Remove Silence, Captions, Generate Clips, Summary and Export.
- Long processes should show progress so the application does not appear frozen.
- Users should be able to safely cancel long tasks where possible.
- Destructive actions should have undo or confirmation.
- The interface should focus on the video preview and editing timeline.

8. Recommended Technology

Area	Recommended Technology
Application Interface	Python + PySide6 desktop application
Video Processing	FFmpeg
Speech-to-Text	Whisper-family transcription
AI Features	LLM-based analysis for clips, titles, summaries and social copy
Data	Local project files and lightweight metadata/database
Performance	Use GPU acceleration when available, with CPU fallback

9. Important Quality Requirements

- **Reliability:** Processing failures should not corrupt projects or original media.
- **Performance:** Basic editing should feel responsive, while long tasks must show progress.

- **Security:** API keys and credentials must not be stored directly in source code.
- **Privacy:** Users should be informed when media is sent to an external AI service.
- **Maintainability:** The interface, video processing, transcription and AI features should be built as separate modules.
- **Data Safety:** Original media files must remain unchanged.

10. How We Will Know the MVP Is Complete

- A user can import a long video and edit it on a timeline.
- The system can detect and remove silence.
- The system can generate synchronized, editable captions.
- The user can choose and customize one of three caption templates.
- The system can export a 1080 × 1920 vertical MP4.
- The original video is never modified.
- The main workflow can be completed by a non-technical user.
- All high-priority MVP features are implemented and tested.

11. Future Improvements

- Speaker detection and support for multiple speakers.
- Automatic scripture or reference detection where relevant.
- Automatic hook and title overlays.
- Intro, outro, logo and lower-third templates.
- Batch processing of an entire folder of videos.
- Automatic publishing integrations, subject to platform permissions.
- Cloud storage and team collaboration.
- Analytics that help improve future AI clip suggestions.

12. Summary

LuxStudio is a focused video-editing platform that combines basic editing tools with AI automation. Its main purpose is to make it easier and faster to turn long recordings into branded, captioned short-form videos. The project should first focus on a stable MVP with editing, silence removal, captions and vertical export before expanding into more advanced AI and collaboration features.

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